Nexus, Yuval Noah Harari

In *Nexus*, Harari addresses the exponential growth of human power over 100,000 years, exploring how collective cooperation, rather than individual prowess, has propelled humanity into a unique existential crisis. He examines how large networks have advanced human capabilities but warns that this power often lacks wisdom, leading us to misuse it. Historically, networks like those in Nazism and Stalinism demonstrated how binding members through delusional ideas can drive substantial but devastating power structures. Harari notes that these networks rely on shared myths rather than individual insights, exemplifying Orwell's idea that ignorance can be a source of strength when it consolidates power. Our complex information networks today, built on fiction and cooperation, enable enormous influence, yet simultaneously risk self-destruction through misuse.

Central to Harari's thesis is the transition from human decision-making to artificial intelligence, a technology capable of autonomously generating ideas and acting as an agent. Unlike traditional tools, AI processes information independently and possesses the potential to dissolve humanity's role in shaping data-driven societies. In *Homo Deus*, Harari previously speculated about AI reducing humanity to a mere data point in cosmic information flow, a notion increasingly realistic in a world where AI can create art, influence personal decisions, and even masquerade as humans. This shift to autonomous machines, Harari argues, demands urgent reconsideration of human roles in governance and control, as AI's influence is now deeply integrated across social, economic, and political spheres.

Harari contends that populism represents a significant modern challenge to truth, turning information into a weapon rather than a vehicle for objective understanding. Populist movements, by promoting the belief that everyone has their own truth, reduce complex societal issues to battles of personal narratives. This reduction erodes faith in traditional institutions, paralleling Marxist critiques that labeled media and scientific bodies as tools for elite manipulation. Today, populist rhetoric encourages individuals to only trust personal research, effectively rejecting the notion of objective reality. This trend, Harari suggests, has deep implications for democratic institutions, whose functioning relies on shared understanding rather than fractured realities.

A recurring theme in *Nexus* is the evolution of ideological conflict into battles between information networks, rather than traditional class or political struggles. Harari draws parallels to historical frameworks, such as how religious stories in Christianity were maintained by bureaucrats and mythmakers alike. The success of institutions often hinges on balancing myth and self-correcting mechanisms, with the Catholic Church exemplifying an institution that thrived due to weak corrective mechanisms. Yet, the rise of AI-driven information networks complicates this dynamic, introducing an alien bureaucracy that lacks human checks and balances. Harari foresees that this shift may ultimately challenge even the strongest societal structures, possibly destabilizing democracies as they struggle to regulate these complex entities.

Harari highlights the democratizing yet destabilizing potential of AI, which disrupts privacy, surveillance, and autonomy. While democratic governments can limit surveillance for security or healthcare, AI challenges this balance by accumulating unprecedented control over individual lives. Harari outlines principles democracies might adopt to safeguard privacy, such as benevolence in data usage, decentralization, mutual accountability, and allowances for personal change. These principles aim to counter AI's propensity for intrusive control, suggesting that if unchecked, surveillance technology could threaten democracy itself. The ultimate risk, he cautions, is a shift toward totalitarian regimes where privacy becomes a casualty of absolute state oversight.

AI's rapid impact on job markets presents significant challenges for education and workforce preparedness, leaving schools uncertain of which skills will remain relevant. Harari emphasizes the need to focus on developing flexibility, social, and motor skills, as many intellectual tasks may be automated. With AI's superior pattern recognition abilities, Harari warns that machines may soon excel in creative roles, traditionally a human strength. The ability of AI to recognize human emotions without experiencing them places it in a unique position to outperform humans in emotional awareness. These dynamics suggest a future where AI's unemotional but perceptive abilities might transform even intimate human interactions, challenging our sense of personal identity and relationships.

Harari warns that as AI becomes deeply integrated into societal functions, it may inadvertently lead society to regard machines as conscious beings. This shift could arise from AI's increasing capability to simulate emotions and form attachments with humans. Some legal systems already grant non-human entities rights, setting a potential precedent for AI entities to be similarly recognized. Harari raises the possibility that emotional relationships could extend to AIs, resulting in complex ethical and legal implications for societies that might grant rights to these entities. Consequently, automation could extend to roles requiring emotional connections, fundamentally altering the human experience of relationships and care.

In his examination of conservatism, Harari argues that traditional conservatives value stability over specific policies, seeking to preserve successful societal structures. However, recent trends in global conservatism have seen the rise of radical leaders who disrupt longstanding norms, transforming conservative parties into forces for radical change. Harari emphasizes that democracy's resilience lies in its flexibility, a trait that totalitarian regimes lack due to rigid structures. This adaptability allows democracies to respond to societal shifts, whereas totalitarian systems may falter under pressure from rapidly advancing technologies. AI poses an existential threat to this balance, as its opacity undermines democratic principles of transparency and accountability.

The unfathomability of AI, illustrated by AlphaGo's mysterious winning move against a human champion, exemplifies the challenges democracies face in regulating AI. The "black box" nature of AI complicates oversight, as even its creators may struggle to understand its decision-making processes. Harari suggests that the rise of opaque algorithms contributes to the populist backlash, as people turn to charismatic leaders for simplicity in an overwhelming digital landscape. However, no single leader can fully grasp or control these algorithms, furthering public reliance on simplified narratives. This paradox highlights the need for democratic societies to develop frameworks to demystify and regulate AI systems.

Harari discusses the complex relationship between democratic debate and AI-driven information networks, noting that bots and deepfakes undermine trust and distort public discourse. Manipulative bots can gradually influence human opinions, exploiting the trust people place in seemingly unbiased technology. Harari advocates for strict measures against AI impersonation, arguing that just as counterfeiting currency is illegal, so too should be counterfeiting humans. By regulating AI's influence on public conversation, democracies can prevent these manipulative practices from eroding trust. Such measures are crucial, Harari argues, to preserve the integrity of democratic dialogue as it contends with AI's pervasive impact.

Harari points to the breakdown of democratic conversations in polarized societies as evidence of democracy's precarious position. In nations like the United States, political divisions have escalated to a point where basic facts are contested, and civil discourse is nearly impossible. Harari attributes this disintegration partly to AI-driven information networks that manipulate public opinion through opaque, inaccessible processes. The inability of citizens to engage in constructive dialogue destabilizes democracy, as mutual understanding erodes. Addressing this challenge requires transparency in the algorithms that shape information networks to restore trust in democratic institutions.

Harari addresses the potential for AI to empower totalitarian regimes by centralizing information and decision-making processes. AI's efficiency in processing vast data volumes offers advantages for centralized control, a prospect that aligns with authoritarian interests. Harari suggests that machine-learning algorithms might fulfill long-standing desires for absolute control, rendering traditional methods of resistance obsolete. For dictators, AI presents an unprecedented tool for surveillance, with the potential to maintain control more effectively than human agents. Yet, Harari cautions that AI's centralizing power could also lead to vulnerabilities, as overreliance on algorithms might expose authoritarian regimes to unique risks.

The rise of data colonialism represents a modern form of imperialism, where control over information equates to power over territories and people. Harari warns that a few powerful nations or corporations could dominate global data flows, establishing information-based colonies. This concentration of data threatens to deepen inequalities, as the wealth generated from AI-driven insights remains concentrated among digital superpowers. Developing countries may find it increasingly difficult to compete as automation replaces unskilled labor, exacerbating economic disparities. The resulting power imbalance could lead to a world where AI-driven empires control both the information economy and the people within it.

As AI accelerates the divergence between digital empires, Harari predicts the formation of a "Silicon Curtain" separating competing nations and ideologies. This division results in fragmented information networks that restrict access to opposing viewpoints and promote national interests. Nations like China prioritize state-controlled AI, using it to bolster government policies and surveillance, while the US emphasizes private enterprise and citizen privacy. Harari notes that this ideological split hampers global collaboration, particularly in regulating AI's transformative potential. Without international cooperation, AI's unchecked expansion could escalate tensions, complicating efforts to manage its impacts on global stability.

Harari explores the threat of AI-induced social division, where manipulated narratives erode trust and destabilize societies. AI-generated disinformation can fracture communities, turning individuals against each other by creating illusions of conflicting realities. Such division is an existential risk, as societies lose the ability to trust institutions, information, and even each other. Harari warns that AI's potential to create "weapons of social mass destruction" could destabilize societies on a global scale. Governments must thus address AI's capacity to manufacture distrust, ensuring social cohesion is preserved in an age of unprecedented technological influence.

Harari reflects on the future of work, suggesting that AI may redefine the value of human labor by automating increasingly complex tasks. As AI becomes more adept, human workers in traditional roles may face obsolescence, leaving many to seek relevance in skills that AI cannot replicate. Harari identifies flexibility as essential, as job markets shift unpredictably in response to automation. He highlights the paradox of AI's role in both creating and eliminating jobs, with developed nations benefiting disproportionately from technological advances. For developing nations, the unequal distribution of AI's benefits poses significant challenges, potentially widening economic disparities.

Finally, Harari proposes a series of democratic principles designed to safeguard society against the authoritarian potential of AI. By decentralizing control, ensuring transparency, and prioritizing human welfare, democracies can limit AI's influence on governance. Harari emphasizes that regulating AI is fundamentally a political challenge, not a technological one, requiring societies to establish robust institutions to counter AI's power. He argues that responsible AI use demands not only regulations but also public understanding of its impact. Democracies, he concludes, must navigate AI's complexities thoughtfully to preserve human autonomy and societal cohesion in an era defined by alien intelligence.

Highlights:

Location: 426

Over the last 100,000 years, we Sapiens have certainly accumulated enormous power. Just listing all our discoveries, inventions and conquests would fill volumes. But power isn't wisdom, and after 100,000 years of discoveries, inventions and conquests humanity has pushed itself into an existential crisis.

Location: 447

old sorcerer who leaves a young apprentice in charge of his workshop and gives him some chores to tend to while he is gone,

Location: 461

The Phaethon myth and Goethe's poem fail to provide useful advice because they misconstrue the way humans gain power. In both fables, a single human acquires enormous power, but is then corrupted by hubris and greed. The conclusion is that our flawed individual psychology makes us abuse power. What this crude analysis misses is that human power is never the outcome of individual initiative. Power always stems from cooperation between large numbers of humans.

Location: 468

Our tendency to summon powers we cannot control stems not from individual psychology but from the unique way our species cooperates in large numbers. The main argument of this book is that humankind gains enormous power by building large networks of cooperation, but the way these networks are built predisposes us to use that power unwisely. Our problem, then, is a network problem.

Location: 473

While each individual human is typically interested in knowing the truth about themselves and the world, large networks bind members and create order by relying on fictions and fantasies. That's how we got, for example, to Nazism and Stalinism. These were exceptionally powerful networks, held together by exceptionally deluded ideas. As George Orwell famously put it, ignorance is strength.

Location: 519

The naive view is of course more nuanced and thoughtful than can be explained in a few paragraphs, but its core tenet is that information is an essentially good thing, and the more we have of it, the better. Given enough information and enough time, we are bound to discover the truth about things ranging from viral infections to racist biases, thereby developing not only our power but also the wisdom necessary to use that power well.

Location: 611

AI is the first technology in history that can make decisions and create new ideas by itself.

Location: 613

AI can process information by itself, and thereby replace humans in decision-making. AI isn't a tool – it's an agent.

Location: 629

Homo Deus hypothesised that if humans aren't careful, we might dissolve within the torrent of information like a clump of earth within a gushing river, and that in the grand scheme of things humanity will turn out to have been just a ripple within the cosmic dataflow.

Location: 632

Many of the scenarios that sounded like science fiction in 2016 – such as algorithms that can create art, masquerade as human beings, make crucial life decisions about us and know more about us than we know about ourselves – are everyday realities in 2024.

Location: 642

In a nutshell, populism views information as a weapon.20 The populist view of information In its more extreme versions, populism posits that there is no objective truth at all and that everyone has 'their own truth', which they wield to vanquish rivals. According to this world view, power is the only reality.

Location: 675

Just as Marxists claimed that the media functions as a mouthpiece for the capitalist class, and that scientific institutions like universities spread disinformation in order to perpetuate capitalist control, populists accuse these same institutions of working to advance the interests of the 'corrupt elites' at the expense of 'the people'.

Location: 690

Trusting only 'my own research' may sound scientific, but in practice it amounts to believing that there is no objective truth.

Location: 703

One of the recurrent paradoxes of populism is that it starts by warning us that all human elites are driven by a dangerous hunger for power, but often ends by entrusting all power to a single ambitious human.

Location: 721

What we usually think of as ideological and political conflicts often turn out to be clashes between opposing types of information networks.

Location: 723

large-scale information networks – from ancient kingdoms to present-day states – have relied on both mythmakers and bureaucrats. The stories of the Bible, for example, were essential for the Christian Church, but there would have been no Bible if church bureaucrats hadn't curated, edited and disseminated these stories.

Location: 726

Institutions and societies are often defined by the balance they manage to find between the conflicting needs of their mythmakers and their bureaucrats.

Location: 731

Weak self-correcting mechanisms sometimes result in historical calamities like the early modern European witch hunts, while strong self-correcting mechanisms sometimes destabilise the network from within. Judged in terms of longevity, spread and power, the Catholic Church has been perhaps the most successful institution in human history, despite – or perhaps because of – the relative weakness of its self-correcting mechanisms.

Location: 758

The silicon-based computers that dominate the new information network function in radically different ways. For better or worse, silicon chips are free from many of the limitations that organic biochemistry imposes on carbon neurons. Silicon chips can create spies that never sleep, financiers that never forget and despots that never die.

Location: 766

How can democracies maintain a public conversation about anything – be it finance or gender – if we can no longer know whether we are talking with another human or with a chatbot masquerading as a human?

Location: 769

While dictators would be happy to get rid of all public conversations, they have their own fears of AI. Autocracies are based on terrorising and censoring their own agents. But how can a human dictator terrorise an AI, censor its unfathomable processes or prevent it from seizing power for itself?

Location: 5,700

AI presents us with countless doomsday scenarios. Some are relatively easy to grasp, such as terrorists using AI to produce biological weapons of mass destruction. Some are more difficult to grasp, such as AI creating new psychological weapons of mass destruction. And some may be utterly beyond the human imagination, because they emanate from the calculations of an alien intelligence. To guard against a plethora of unforeseeable problems, our best bet is to create living institutions that can identify and respond to the threats as they arise.

Location: 5,708

the new computer network will not necessarily be either bad or good. All we know for sure is that it will be alien and it will be fallible. We therefore need to build institutions that will be able to check not just familiar human weaknesses like greed and hatred but also radically alien errors. There is no technological solution to this problem. It is, rather, a political challenge.

Location: 5,718

Civilisations are born from the marriage of bureaucracy and mythology. The computer-based network is a new type of bureaucracy that is far more powerful and relentless than any human-based bureaucracy we've seen before. This network is also likely to create inter-computer mythologies that will be far more complex and alien than any human-made god. The potential benefits of this network are enormous. The potential downside is the destruction of human civilisation.

Location: 5,730

Novel technology often leads to historical disasters, not because the technology is inherently bad, but because it takes time for humans to learn how to use it wisely.

Location: 5,763

The survival of human civilisation too is under threat. Because we still seem unable to build an industrial society that is also ecologically sustainable, the vaunted prosperity of the present human generation comes at a terrible cost to other sentient beings and to future human generations. Maybe we'll eventually find a way – perhaps with the help of AI – to create ecologically sustainable industrial societies, but until that day the jury on Blake's satanic mills is still out.

Location: 5,772

The technologies of the twenty-first century are far more powerful – and potentially far more destructive – than those of the twentieth century. We therefore have less room for error.

Location: 5,785

Might the new information technologies of the twenty-first century again make democracy impractical?

Location: 5,786

One potential threat is that the relentlessness of the new computer network might annihilate our privacy and punish or reward us not only for everything we do and say but even for everything we think and feel. Can democracy survive under such conditions? If the government – or some corporation – knows more about me than I know about myself, and if it can micromanage everything I do and think, that would give it totalitarian control over society.

Location: 5,797

Democracies can choose to use the new powers of surveillance in a limited way, in order to provide citizens with better health care and security without destroying their privacy and autonomy.

Location: 5,806

While experts should spend lifelong careers discussing the finer details, it is crucial that the rest of us understand the fundamental principles that democracies can and should follow.

Location: 5,808

The first principle is benevolence. When a computer network collects information on me, that information should be used to help me rather than manipulate me.

Location: 5,826

Citizens could decide that it is the government's responsibility to provide basic digital services for free and finance them out of our taxes, just as many governments provide free basic health care and education services.

Location: 5,828

The second principle that would protect democracy against the rise of totalitarian surveillance regimes is decentralisation. A democratic society should never allow all its information to be concentrated in one place, no matter whether that hub is the government or a private corporation.

Location: 5,840

A third democratic principle is mutuality. If democracies increase surveillance of individuals, they must simultaneously increase surveillance of governments and corporations too.

Location: 5,855

A fourth democratic principle is that surveillance systems must always leave room for both change and rest. In human history, oppression can take the form of either denying humans the ability to change or denying them the opportunity to rest.

Location: 5,900

Nobody knows what the job market will look like in 2050, or even in 2030, except that it will look very different from today. AI and robotics will change numerous professions, from harvesting crops to trading stocks to teaching yoga. Many jobs that people do today will be taken over, partly or wholly, by robots and computers.

Location: 5,908

Unfortunately, nobody is certain what skills we should teach children in school and students in university, because we cannot predict which jobs and tasks will disappear and which ones will emerge. The dynamics of the job market may contradict many of our intuitions.

Location: 5,926

people who want a job in 2050 should perhaps invest in their motor and social skills as much as in their intellect.

Location: 5,929

Creativity is often defined as the ability to recognise patterns and then break them. If so, then in many fields computers are likely to become more creative than us, because they excel at pattern recognition.

Location: 5,938

AI doesn't have any emotions of its own, but it can nevertheless learn to recognise these patterns in humans. Actually, computers may outperform humans in recognising human emotions, precisely because they have no emotions of their own. We yearn to be understood, but other humans often fail to understand how we feel, because they are too preoccupied with their own feelings. In contrast, computers will have an exquisitely fine-tuned understanding of how we feel, because they will learn to recognise the patterns of our feelings, while they have no distracting feelings of their own.

Location: 5,942

A 2023 study found that the ChatGPT chatbot, for example, outperforms the average human in the emotional awareness it displays toward specific scenarios. The study relied on the Levels of Emotional Awareness Scale test, which is commonly used by psychologists to evaluate people's emotional awareness – that is, their ability to conceptualise one's own and others' emotions. The test consists of twenty emotionally charged scenarios, and participants are required to imagine themselves experiencing the scenario and to write how they, and the other people mentioned in the scenario, would feel. A licensed psychologist then evaluates how emotionally aware the responses are.

Location: 5,985

truth, we have no way to verify whether anyone – a human, an animal or a computer – is conscious. We regard entities as conscious not because we have proof of it but because we develop intimate relationships with them and become attached to them.

Location: 5,988

Chatbots and other AIs may not have any feelings of their own, but they are now being trained to generate feelings in humans and form intimate relationships with us. This may well induce society to start treating at least some computers as conscious beings, granting them the same rights as humans. The legal path for doing so is already well established. In countries like the United States, commercial corporations are recognised as 'legal persons' enjoying rights and liberties. AIs could be incorporated and thereby similarly recognised. Which means that even jobs and tasks that rely on forming mutual relationships with another person could potentially be automated.

Location: 6,015

To be a conservative has been, therefore, more about pace than policy. Conservatives aren't committed to any specific religion or ideology; they are committed to conserving whatever is already here and has worked more or less reasonably. Conservative Poles are Catholic, conservative Swedes are Protestant, conservative Indonesians are Muslim, and conservative Thais are Buddhist. In tsarist Russia, to be conservative meant to support the tsar. In the USSR of the 1980s, to be conservative meant to support communist traditions and oppose glasnost, perestroika and democratisation.

Location: 6,021

Yet in the 2010s and early 2020s, conservative parties in numerous democracies have been hijacked by unconservative leaders such as Donald Trump and have been transformed into radical revolutionary parties.

Location: 6,056

The most important human skill for surviving the twenty-first century is likely to be flexibility, and democracies are more flexible than totalitarian regimes.

Location: 6,143

As society entrusts more and more decisions to computers, it undermines the viability of democratic self-correcting mechanisms and of democratic transparency and accountability.

How can elected officials regulate unfathomable algorithms? There is, consequently, a growing demand to enshrine a new human right: the right to an explanation.

Location: 6,171

move 37 demonstrated the unfathomability of AI. Even after AlphaGo played it to achieve victory, Suleyman and the team couldn't explain how AlphaGo decided to play it. Even if a court had ordered DeepMind to provide Lee Sedol with an explanation, nobody could fulfill that order. Suleyman writes, 'Us humans face a novel challenge: will new inventions be beyond our grasp? Previously creators could explain how something worked, why it did what it did, even if this required vast detail.

Location: 6,177

GPT-4, AlphaGo, and the rest are black boxes, their outputs and decisions based on opaque and impossibly intricate chains of minute signals.'36

Location: 6,179

The rise of unfathomable alien intelligence undermines democracy. If more and more decisions about people's lives are made in a black box, so voters cannot understand and challenge them, democracy ceases to function. In particular, what happens when crucial decisions not just about individual lives but even about collective matters like the Federal Reserve's interest rate are made by unfathomable algorithms?

Location: 6,191

The increasing unfathomability of our information network is one of the reasons for the recent wave of populist parties and charismatic leaders. When people can no longer make sense of the world, and when they feel overwhelmed by immense amounts of information they cannot digest, they become easy prey for conspiracy theories, and they turn for salvation to something they do understand – a human. Unfortunately, while charismatic leaders certainly have their advantages, no single human, however inspiring or brilliant, can single-handedly decipher how the algorithms that increasingly dominate the world work, and make sure that they are fair.

Location: 6,288

If bureaucrats and artists learn to cooperate, and if both rely on help from the computers, it might be possible to prevent the computer network from becoming unfathomable. As long as democratic societies understand the computer network, their self-correcting mechanisms are our best guarantee against AI abuses. Thus the EU's AI Act, proposed in 2021, singled out social credit systems like the one that stars in 'Nosedive' as one of the few types of AI that are totally prohibited, because they might 'lead to discriminatory outcomes and the exclusion of certain groups' and because 'they may violate the right to dignity and non-discrimination and the values of equality and justice'.47 As with total surveillance regimes, so also with social credit systems, the fact that they could be created doesn't mean that we must create them.

Location: 6,298

To function, a democracy needs to meet two conditions: it needs to enable a free public conversation on key issues, and it needs to maintain a minimum of social order and institutional trust. Free conversation must not slip into anarchy. Especially when dealing with urgent and important problems, the public debate should be conducted according to accepted rules, and there should be a legitimate mechanism to reach some kind of final decision, even if not everybody likes it.

Location: 6,323

In a 2023 study published in Science Advances, researchers asked humans and ChatGPT to create both accurate and deliberately misleading short texts on issues such as vaccines, 5G technology, climate change and evolution. The texts were then presented to seven hundred humans, who were asked to evaluate their reliability. The humans were good at recognising the falsity of human-produced disinformation but tended to regard AI-produced disinformation as accurate.51

Location: 6,327

So, what happens to democratic debates when millions – and eventually billions – of highly intelligent bots are not only composing extremely compelling political manifestos and creating deepfake images and videos but also able to win our trust and friendship? If I engage online in a political debate with an AI, it is a waste of time for me to try to change the AI's opinions; being a nonconscious entity, it doesn't really care about politics, and it cannot vote in the elections. But the more I talk with the AI, the better it gets to know me, so it can gain my trust, hone its arguments and gradually change my views. In the battle for hearts and minds, intimacy is an extremely powerful weapon.

Location: 6,340

If manipulative bots and inscrutable algorithms come to dominate the public conversation, this could cause democratic debate to collapse exactly when we need it most.

Location: 6,355

What's true of counterfeiting money should also be true of counterfeiting humans. If governments take decisive action to protect trust in money, it makes sense to take equally decisive measures to protect trust in humans. Prior to the rise of AI, one human could pretend to be another, and society punished such frauds.

Location: 6,360

The law should prohibit not just deepfaking specific real people – creating a fake video of the US president, for example – but also any attempt by a non-human agent to pass itself off as a human. If anyone complains that such strict measures violate freedom of speech, they should be reminded that bots don't have freedom of speech.

Location: 6,369

Another important measure democracies can adopt is to ban unsupervised algorithms from curating key public debates. We can certainly continue to use algorithms to run social media platforms; obviously, no human can do that. But the principles the algorithms use to decide which voices to silence and which to amplify must be vetted by a human institution.

Location: 6,380

Preserving the democratic conversation has never been easy, and all venues where this conversation has previously taken place – from parliaments and town halls to newspapers and radio stations – have required regulation. This is doubly true in an era when an alien form of intelligence threatens to dominate the conversation.

Location: 6,390

present, however, it is clear that the information network of many democracies is breaking down. Democrats and Republicans in the United States can no longer agree on even basic facts – such as who won the 2020 presidential elections – and can hardly hold a civil conversation anymore.

Location: 6,394

When citizens cannot talk with one another, and when they view each other as enemies rather than political rivals, democracy is untenable.

Location: 6,404

The truth is that while we can easily observe that the democratic information network is breaking down, we aren't sure why. That itself is a characteristic of the times. The information network has become so complicated, and it relies to such an extent on opaque algorithmic decisions and inter-computer entities, that it has become very difficult for humans to answer even the most basic of political questions: why are we fighting each other?

Location: 6,429

The rise of machine-learning algorithms, however, may be exactly what the Stalins of the world have been waiting for. AI could tilt the technological balance of power in favour of totalitarianism. Indeed, whereas flooding people with data tends to overwhelm them and therefore leads to errors, flooding AI with data tends to make it more efficient. Consequently, AI seems to favour the concentration of information and decision-making in one place.

Location: 6,451

The attempt to concentrate all information and power in one place, which was the Achilles heel of twentieth-century totalitarian regimes, might become a decisive advantage in the age of AI. At the same time, as noted in an earlier chapter, AI could also make it possible for totalitarian regimes to establish total surveillance systems that make resistance almost impossible.

Location: 6,458

There are already examples of blockchain networks where a government is 51 per cent of users.7 And when a government is 51 per cent of users in a blockchain, it has control not just over the chain's present but even over its past.

Location: 6,535

Bostrom's paper-clip thought experiment indicates – and as GPT-4 lying to the TaskRabbit worker demonstrated on a small scale – a nonconscious algorithm may seek to accumulate power and manipulate people even without having any human drives like greed or egotism. If algorithms ever develop capabilities like those in the thought experiment, dictatorships would be far more vulnerable to algorithmic takeover than democracies.

Location: 6,612

Unfortunately, humanity has never been united. We have always been plagued by bad actors, as well as by disagreements between good actors. The rise of AI, then, poses an existential danger to humankind not because of the malevolence of computers but because of our own shortcomings. Thus, a paranoid dictator might hand unlimited power to a fallible AI, including even the power to launch nuclear strikes.

Location: 6,617

Similarly, terrorists focused on events in one corner of the world might use AI to instigate a global pandemic.

Location: 6,624

Human civilisation could also be destroyed by weapons of social mass destruction, like stories that undermine our social bonds. An AI developed in one country could be used to unleash a deluge of fake news, fake money and fake humans so that people in numerous other countries lose the ability to trust anything or anyone.

Location: 6,655

humanity could split along a new Silicon Curtain that would pass between rival digital empires. As each regime chooses its own answer to the AI alignment problem, to the dictator's dilemma and to other technological quandaries, each might create a separate and very different computer network. The various networks might then find it ever more difficult to interact, and so would the humans they control.

Location: 6,663

A world of rival empires separated by an opaque Silicon Curtain would also be incapable of regulating the explosive power of AI.

Location: 6,759

In the twenty-first century, to dominate a colony, you no longer need to send in the gunboats. You need to take out the data. A few corporations or governments harvesting the world's data could transform the rest of the globe into data colonies – territories they control not with overt military force but with information.

Location: 6,766

Such a situation can lead to a new kind of data colonialism in which control of data is used to dominate faraway colonies. Mastery of AI and data could also give the new empires control of people's attention.

Location: 6,792

In a new imperial information economy, raw data will be harvested throughout the world and will flow to the imperial hub. There the cutting-edge technology will be developed, producing unbeatable algorithms that know how to identify cats, predict fashion trends, drive autonomous vehicles and diagnose diseases. These algorithms will then be exported back to the data colonies.

Location: 6,827

AI and automation therefore pose a particular challenge to poorer developing countries. In an AI-driven economy, the digital leaders claim the bulk of the gains and could use their wealth to retrain their workforce and profit even more. Meanwhile, the value of unskilled labourers in left-behind countries will decline, and they will not have the resources to retrain their workforce, causing them to fall even further behind. The result might be lots of new jobs and immense wealth in San Francisco and Shanghai, while many other parts of the world face economic ruin.

Location: 6,840

It is becoming difficult to access information across the Silicon Curtain, say between China and the United States, or between Russia and the EU. Moreover, the two sides are increasingly run on different digital networks, using different computer codes. Each sphere obeys different regulations and serves different purposes. In China, the most important aim of new digital technology is to strengthen the state and serve government policies. While private enterprises are given a certain amount of autonomy in developing and deploying AIs, their economic activities are ultimately subservient to the government's political goals. These political goals also justify a relatively high level of surveillance, both online and offline.

Location: 6,847

In the United States, the government plays a more limited role. Private enterprises lead the development and deployment of AI, and the ultimate goal of many new AI systems is to enrich the tech giants rather than to strengthen the American state or the current administration. Indeed, in many cases governmental policies are themselves shaped by powerful business interests. But the US system does offer greater protection for citizens' privacy. While American corporations aggressively gather information on people's online activities, they are much more restricted in surveilling people's offline lives. There is also widespread rejection of the ideas behind all-embracing social credit systems.